



University of Stuttgart
Collaborative
Artificial Intelligence



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European Laboratory for Learning and Intelligent Systems



University of Stuttgart
Germany

Machine Perception and Learning for Collaborative Intelligent Systems

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www.collaborative-ai.org 

Generative Models I

- Mixture of Gaussians
- Autoencoders
- Variational Autoencoders (VAEs)

Variational Autoencoders: Recap

Generative Models II

Generative Adversarial Networks (GANs)

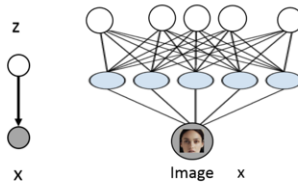
Extensions

Applications

Case Study

- Probabilistic version of autoencoders
- Allows for sampling from the learned model to generate new, unseen samples

- Put a prior on z : $z \sim \mathcal{N}(0, I)$



Source: Paul Liang, CMU

$p(x|z) = \mathcal{N}(\mu_{\theta}(z), \Sigma_{\theta}(z))$ where μ_{θ} and Σ_{θ} are neural networks

$$\begin{aligned}
 \mathcal{L}(\mathbf{x}; \theta, \phi) &= E_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log q_{\phi}(\mathbf{z}|\mathbf{x})] \\
 &= E_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{z}, \mathbf{x}; \theta) - \log p(\mathbf{z}) + \log p(\mathbf{z}) - \log q_{\phi}(\mathbf{z}|\mathbf{x})] \\
 &= E_{q_{\phi}(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x}|\mathbf{z}; \theta)] - D_{KL}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))
 \end{aligned}$$

reconstruction
prior

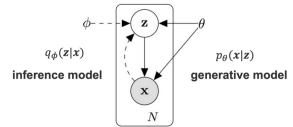
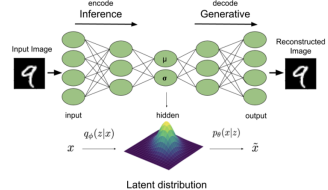


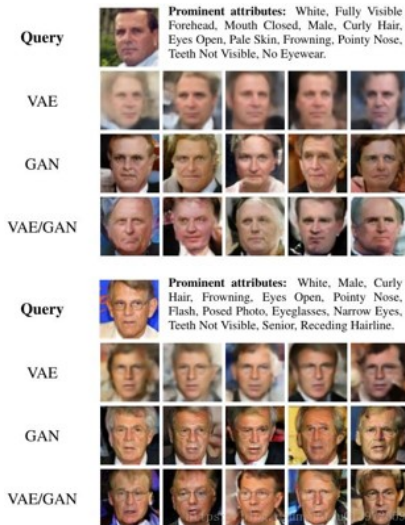
Figure courtesy: Kingma & Welling, 2014

1. Take a datapoint $\mathbf{x}_i$.
2. Map it to μ, σ using $q_{\phi}(\mathbf{z}|\mathbf{x}_i)$. **encoder**
3. Sample $\epsilon \sim N(0, I)$ and compute $\hat{\mathbf{z}} = \mu + \sigma\epsilon$. **reparameterize**
4. Reconstruct $\hat{\mathbf{x}}$ by sampling from $p(\mathbf{x}|\hat{\mathbf{z}}; \theta)$. **decoder**

Differentiable using reparameterization trick



Source: Paul Liang, CMU



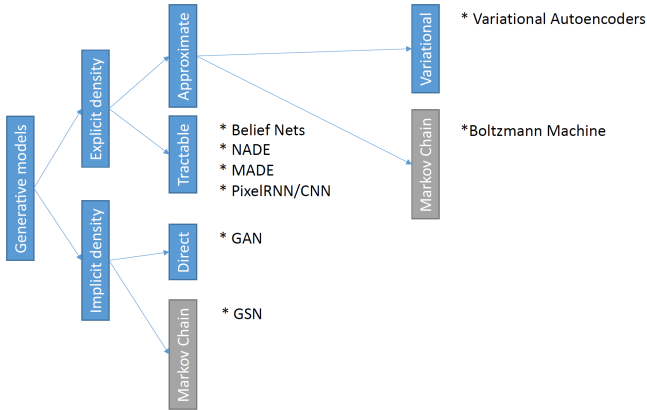
Source: Paul Liang, CMU

VAEs define intractable density function with latent z :

$$p_{\theta}(x) = \int p_{\theta}(z)p_{\theta}(x|z)dz$$

Cannot optimise directly, derive and optimise lower bound on likelihood instead

- Relatively easier to train
- Explicit inference network $q(z|x)$
- More blurry images (due to reconstruction)



Source: Otmar Hilliges and Ian Goodfellow

What if we give up on explicitly modelling density, and just want ability to sample?

Variational Autoencoders: Recap

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Generative Models II

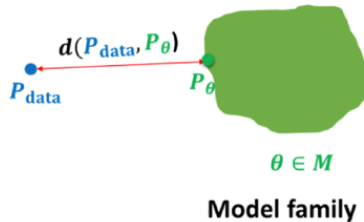
Generative Adversarial Networks (GANs)

- Paul Liang and Louis-Philippe Morency @CMU
- Fei-Fei Li, Justin Johnson, and Serena Yeung @Stanford
- O. Hilliges @ETHZ

- Beyond likelihood-based learning:
 - $p(x)$ difficult to evaluate and optimise in high-dimensions
 - High $p(x)$ may not correspond to realistic samples

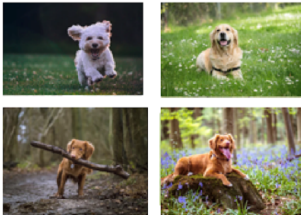


$x_i \sim P_{\text{data}}$
 $i = 1, 2, \dots, n$



Source: Paul Liang @ CMU

- Towards likelihood-free learning



$$S_1 = \{\mathbf{x} \sim P\}$$

vs.



$$S_2 = \{\mathbf{x} \sim Q\}$$

Source: Paul Liang @ CMU

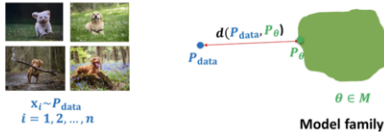
- Given a finite set of samples from two distributions, how can we tell if they belong to the same distribution? i.e. $P = Q$

- Given $S_1 = x \sim P$ and $S_2 = x \sim Q$, a two-sample test considers the following hypothesis
 - Null hypothesis $H_0 : P = Q$
 - Alternate hypothesis: $H_1 : P \neq Q$

- The test T compares the differences between S_1 and S_2 in terms of means and variance
- If T is less than α , then we accept H_0

Key observation: Test statistic is **likelihood-free** since it does not involve the densities P or Q , only samples

- Towards likelihood-free learning

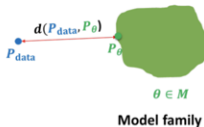


Source: Paul Liang @ CMU

- Assume we have access to $S_1 = D = \{x \sim p_{\text{data}}\}$
- We also have our model's distribution p_{θ}
- Assume that our model's distribution permits efficient sampling of $S_2 = \{x \sim p_{\theta}\}$



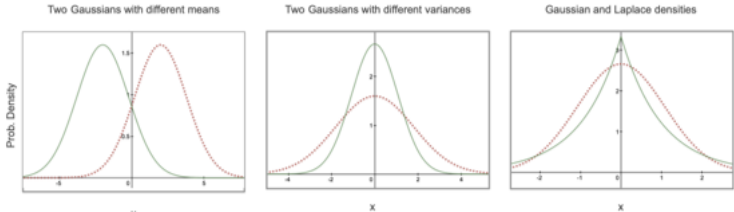
$\mathbf{x}_i \sim P_{\text{data}}$
 $i = 1, 2, \dots, n$



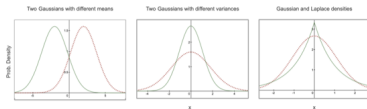
Source: Paul Liang @ CMU

- **Alternate notion of distance between distributions:** Train the generative model to minimise a two-sample test objective between S_1 and S_2
- **Key observation:** Test statistic is **likelihood-free** since it does not involve the densities P or Q , only samples

- Towards likelihood-free learning



Source: Paul Liang @ CMU



Source: Paul Liang @ CMU

- Towards likelihood-free learning
- Problem: finding a two-sample test objective in high-dimensions is hard
- In the generative model setup, we know that S_1 and S_2 come from different distributions
- Key idea: learn a statistic that maximises a suitable notion of distance between the two sets of samples S_1 and S_2 : p_{data} and p_{θ}

- Problem: Want to sample from complex, high-dimensional training distribution. No direct way to do this!

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- Solution: Sample from a simple distribution, e.g. random noise. Learn transformation to training distribution.

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- Solution: Sample from a simple distribution, e.g. random noise. Learn transformation to training distribution.
- What can we use to represent this **complex** transformation?
A neural network!

Output: Sample from
training distribution



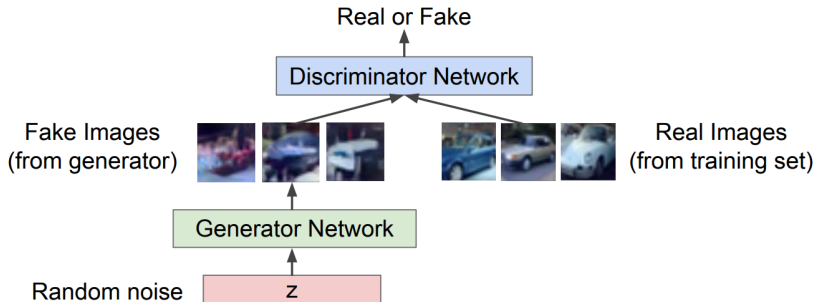
Generator
Network

Input: Random noise

z

Source: Fei-Fei Li & Justin Johnson & Serena Yeung @ Stanford

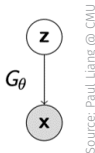
- **Generator network:** try to fool the discriminator by generating real-looking images.
- **Discriminator network:** try to distinguish between real and fake images.



Source: Fei-Fei Li & Justin Johnson & Serena Yeung @ Stanford. Images originally from Denton et al. [2015]

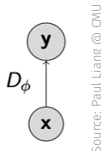
Vanilla GANs proposed by Goodfellow et al. [2014]

- A two-player minimax game between a generator and a discriminator



- **Generator:** a directed latent variable model from z to x
- Minimises the two-sample test objective: in support of the null hypothesis $p_{data} = p_\theta$

- A two-player minimax game between a generator and a discriminator



Source: Paul Liang @ CMU

- **Discriminator:** any function (e.g. neural network) that tries to distinguish “real” samples from the datasets from “fake” samples generated by the model
- Maximises the two-sample test objective: in support of the alternative hypothesis $p_{data} \neq p_\theta$

Slide source: [Goodfellow et al., 2014]

- Training objective for **discriminator**

$$\max_D V(G, D) = E_{x \sim p_{data}} [\log D(x)] + E_{x \sim p_G} [\log(1 - D(x))]$$

- For a fixed generator G , the discriminator performs binary classification between true samples (assign label 1) vs fake samples (assign label 0)
- Optimal discriminator:

$$D_G^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)}$$

- Training objective for **generator**

$$\min_G V(G, D) = E_{x \sim p_{data}} [\log D(x)] + E_{x \sim p_G} [\log(1 - D(x))]$$

- For the optimal generator $D_G^*(\cdot)$ we have:

$$V(F, D_G^*(x)) =$$

$$= E_{x \sim p_{data}} \left[\frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \right] + E_{x \sim p_G} \left[\frac{p_G(x)}{p_{data}(x) + p_G(x)} \right] =$$

$$= E_{x \sim p_{data}} \left[\frac{p_{data}(x)}{\frac{p_{data}(x) + p_G(x)}{2}} \right] + E_{x \sim p_G} \left[\frac{p_G(x)}{\frac{p_{data}(x) + p_G(x)}{2}} \right] - \log 4 =$$

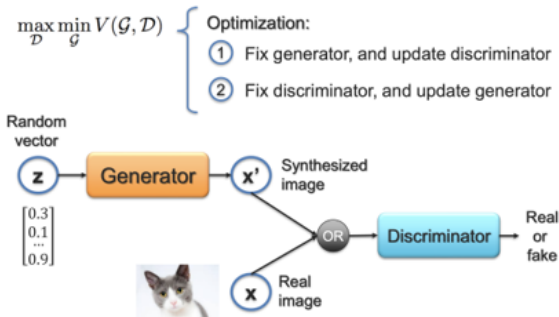
$$= D_{KL}[p_{data}, \frac{p_{data} + p_G}{2}] + D_{KL}[p_G, \frac{p_{data} + p_G}{2}] - \log 4 =$$

$$= 2D_{JSD}[p_{data}, p_G] - \log 4$$

- Also known as the symmetric KL divergence

$$D_{JSD} = \frac{1}{2}(D_{KL}[p, \frac{p+q}{2}] + D_{KL}[q, \frac{p+q}{2}])$$

- Properties:
 - $D_{JSD}[p, q] \geq 0$
 - $D_{JSD}[p, q] = 0$ iff $p = q$
 - $D_{JSD}[p, q] = D_{JSD}[q, p]$
 - $\sqrt{D_{JSD}[p, q]}$ satisfies the triangle inequality $\rightarrow$
Jensen-Shannon Distance
- Optimal generator for the JSD/Negative Cross Entropy GAN



Source: Paul Liang @ CMU

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

for k steps **do**

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log \left(1 - D(G(z^{(i)})) \right) \right].$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by descending its stochastic gradient:

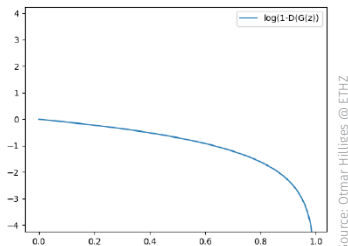
$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(1 - D(G(z^{(i)})) \right).$$

end for

Source: Otmar Hilliges @ ETHZ

Objective:

$$\min_{\Theta_g} \max_{\Theta_d} \mathbb{E}_{x \sim p_d(x)} [\log D_{\Theta_d}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\Theta_d}(G_{\Theta_g}(z)))]$$



Source: Otmar Hilliges @ ETHZ

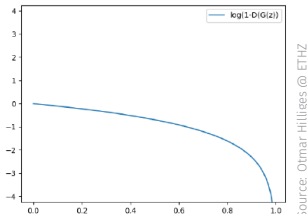
Alternate between:

1. Gradient *ascent* on D:

$$\max_{\Theta_d} \mathbb{E}_{x \sim p_d(x)} [\log D_{\Theta_d}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\Theta_d}(G_{\Theta_g}(z)))]$$

2. Gradient *descent* on G:

$$\min_{\Theta_g} \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\Theta_d}(G_{\Theta_g}(z)))]$$



- In practice, optimising this generator objective does not work well!
- When sample is likely fake, we want to learn from it to improve generator. But gradient in this region is relatively flat!
- Gradient signal dominated by region where sample is already good

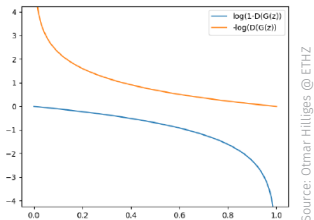
Alternate between:

1. Gradient *ascent* on D:

$$\max_{\Theta_d} \mathbb{E}_{x \sim p_d(x)} [\log D_{\Theta_d}(x)] + \mathbb{E}_{z \sim p_z(z)} [\log(1 - D_{\Theta_d}(G_{\Theta_g}(z)))]$$

2. **Instead**: gradient **ascent** on G:

$$\max_{\Theta_g} \mathbb{E}_{z \sim p_z(z)} [\log(D_{\Theta_d}(G_{\Theta_g}(z)))]$$



- Instead of minimizing likelihood of discriminator being correct, now maximize likelihood of discriminator being wrong.
- Same objective of fooling discriminator, but now higher gradient signal for bad samples → works much better! Standard in practice.

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k , is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

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- Sample minibatch of m examples $\{x^{(1)}, \dots, x^{(m)}\}$ from data generating distribution $p_{\text{data}}(x)$.
- Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D(x^{(i)}) + \log (1 - D(G(z^{(i)}))) \right].$$

end for

- Sample minibatch of m noise samples $\{z^{(1)}, \dots, z^{(m)}\}$ from noise prior $p_g(z)$.
- Update the generator by ascending its stochastic gradient (improved objective):

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(D_{\theta_d} \left(G_{\theta_g}(z^{(i)}) \right) \right)$$

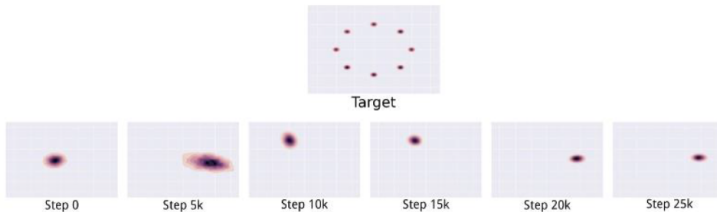
end for

Source: Omar Hilliges @ ETHZ

- Jointly training two networks is challenging, can be unstable. Choosing objectives with better loss landscapes helps training, is an active area of research.
- Some find $k = 1$ more stable. Others use $k > 1$, no best rule.
- Other GAN variants alleviate this problem and are more stable.

- Difficult to train.
- Need to find Nash Equilibrium in two player game. Making downhill progress for one player may move the other player uphill.
- Generator can trick discriminator due to underlying image statistics.

- Difficult to train
- Mode collapse (no sample diversity): saddle point in dual energy landscape



Source: Otmar Hilliges @ ETHZ

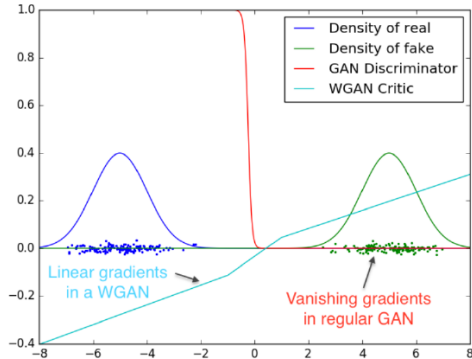
Unrolled Generative Adversarial Networks [Metz et al., 2017]

- Implicit pdf in VAE with intractable data likelihood
- No variational bound is needed
- GANs only care about samples
- Sharper images

- JSD correlates badly with sample quality → do not know when to stop training
- D tends to be optimal, JSD saturates → bad gradients
- One solution: Use Wasserstein distance

GAN objective can be generalized to entire family of divergences:

f-GAN: Training Generative Neural Samplers using Variational Divergence Minimization [Nowozin et al., 2016]



Source: Otmar Hilliges @ ETHZ

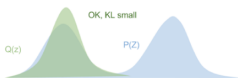
Wasserstein GANs [Arjovsky et al., 2017]

Generative Models II

Extensions

$$D_{JSD} = \frac{1}{2} (D_{KL}[p, \frac{p+q}{2}] + D_{KL}[q, \frac{p+q}{2}])$$

- If our data are on a low-dimensional manifold of a high-dimensional space, the model's manifold and the true data manifold can have a negligible intersection in practice
- KL divergence is undefined or infinite so the loss function and gradients may not be continuous and well behaved

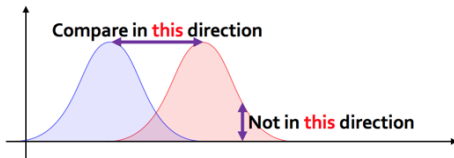
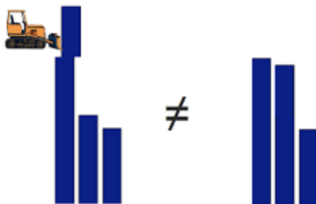


Source: [Arjovsky et al., 2017]

$$W(P||Q) = \inf_{\gamma \in \Pi(P,Q)} E_{(x,y) \sim \gamma} [\|x - y\|]$$

- $\Pi(P, Q)$ denotes the set of all joint distributions $\gamma(x, y)$ whose marginals are P and Q
- $\gamma(x, y)$ indicates a plan to transport “mass” from x to y , when deforming P into Q
- The Wasserstein or Earth Mover’s Distance is then the “cost” of the **optimal** transport plan

$$W(P||Q) = \inf_{\gamma \in \Pi(P,Q)} E_{(x,y) \sim \gamma} [\|x - y\|]$$



$$W(P||Q) = \inf_{\gamma \in \Pi(P,Q)} E_{(x,y) \sim \gamma} [\|x - y\|]$$

- The Wasserstein Distance is well defined
- Earth Mover's Distance: minimum transportation cost for making one pile of dirt in the shape of one probability distribution to the shape of the other distribution

Generative Models II

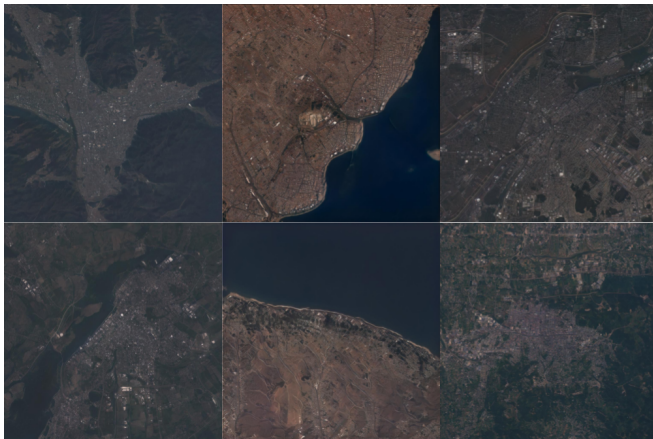
Applications

This person does not exist



Source: <https://this-person-does-not-exist.com/en>

This city does not exist



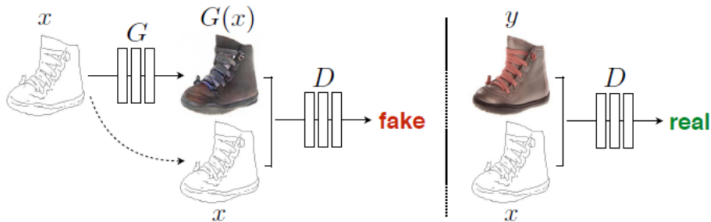
Source: <https://https://thiscitydoesnotexist.com/>

Want to explore different application domains?

- *<https://thisxdoesnotexist.com/>* - A comprehensive list
- *<https://imgflip.com/ai-meme>* - Memes
- *<https://thisartworkdoesnotexist.com/>* - Artworks
- *<https://www.thisautomobiledoesnotexist.com/>* - Cars

and many more...

Image-to-Image Translation with Conditional Adversarial Networks



Source: Otmar Hilliges @ ETHZ

Slide source: [Isola et al., 2017]

Image-to-Image Translation with Conditional Adversarial Networks

The Objective is: $\mathcal{L}(G, D) = \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G)$

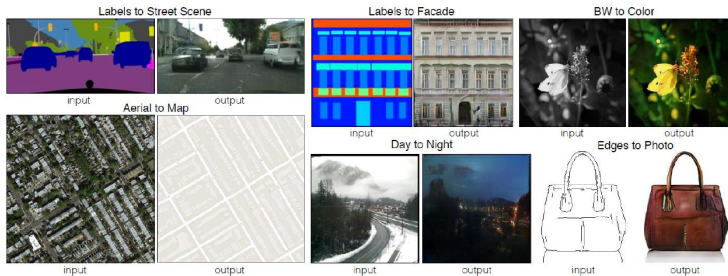
where $\mathcal{L}_{cGAN}$ is a conditional GAN objective:

$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x,y}[\log D(x, y)] + \mathbb{E}_{x,z}[1 - \log D(x, G(x, z))]$$

and $\mathcal{L}_{L1}$:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z}[\|y - G(x, z)\|_1]$$

Image-to-Image Translation with Conditional Adversarial Networks



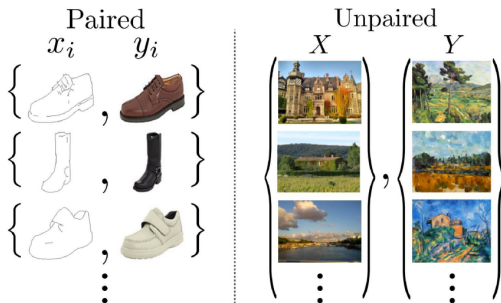
Source: Otmar Hilliges @ ETHZ

Slide source: [Isola et al., 2017]

- One obvious limitation is that this approach requires paired images as training data
- In many cases, one has at its disposal examples from two densities and wants to translate a sample from the first (“images of apples”) into a sample likely under the second (“images of oranges”).

Slide source: [Isola et al., 2017]

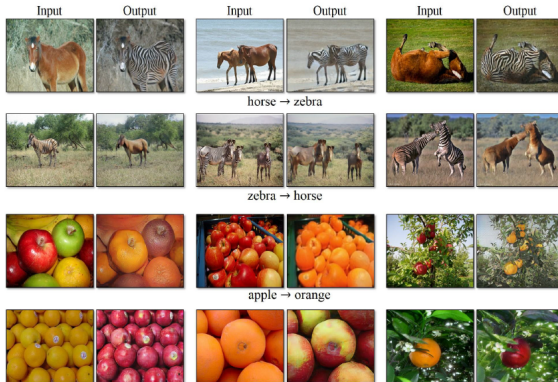
Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks



Source: Otmar Hilliges @ ETHZ

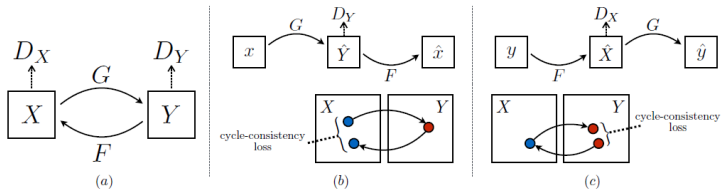
Slide source: [Zhu et al., 2017]

Unpaired Image-to-Image Translation using Cycle-Consistent Adversarial Networks



Source: Otmar Hilliges @ ETHZ

Slide source: [Zhu et al., 2017]



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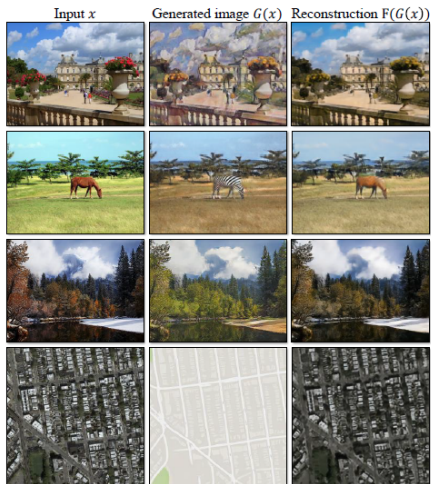
The objective is:

$$\mathcal{L}(G, F, D_X, D_Y) = \mathcal{L}_{GAN}(G, D_Y, X, Y) + \mathcal{L}_{GAN}(F, D_X, Y, X) + \lambda \mathcal{L}_{cyc}(G, F)$$

where $\mathcal{L}_{GAN}$ is the standard GAN loss as previously and $\mathcal{L}_{cyc}$ is:

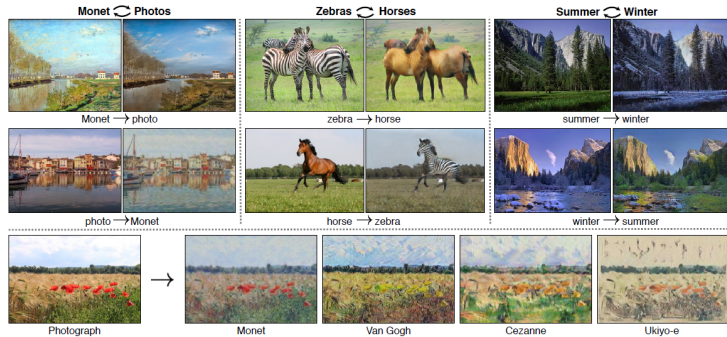
$$\mathcal{L}_{cyc} = \mathbb{E}_{x \sim p_{data}(x)} [\|F(G(x)) - x\|_1] + \mathbb{E}_{y \sim p_{data}(y)} [\|G(F(y)) - y\|_1]$$

Slide source: [Zhu et al., 2017]



Source: Otmar Hilliges @ ETHZ

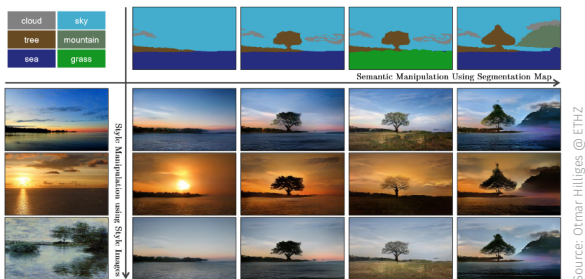
Slide source: [Zhu et al., 2017]



Source: Otmar Hilliges @ ETHZ

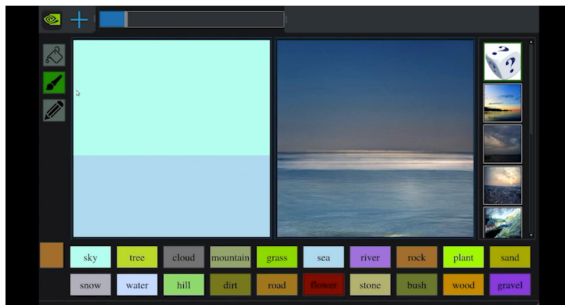
Slide source: [Zhu et al., 2017]

- Semantic Image Synthesis with Spatially-Adaptive Normalization
- Task: Given semantic segmentation and reference style, synthesise image



Slide source: [Park et al., 2019]

Semantic image synthesis with spatially-adaptive normalization

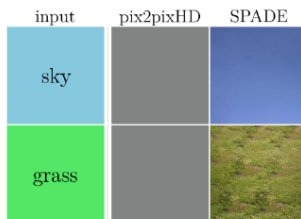
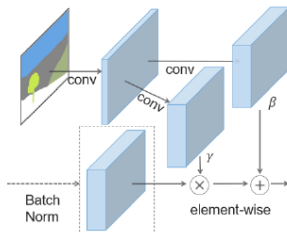


Source: Otmar Hilliges @ ETHZ

Slide source: [Park et al., 2019]

Semantic image synthesis with spatially-adaptive normalization

- Unconditional normalisation layer in generator “washes away” information of semantic labels
- Proposal: Spatially-varying conditional normalisation

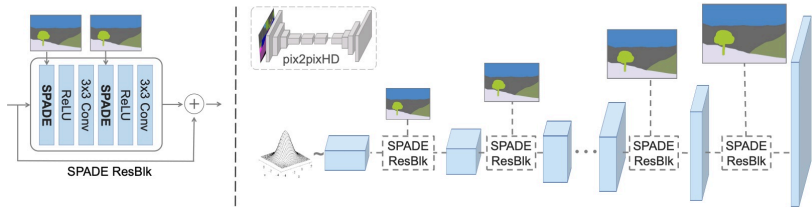


Source: Semantic Image Synthesis With Spatially-Adaptive Normalization

Slide source: [Park et al., 2019]

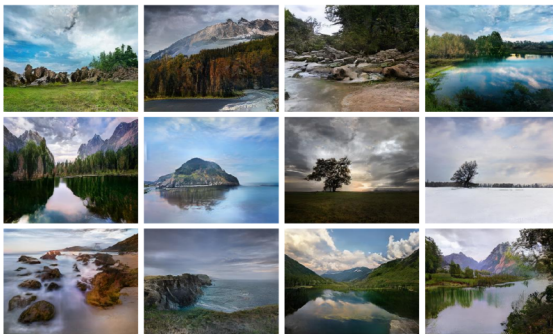
- The mask is first projected onto an embedding space.
- Then convolved to produce the modulation parameters γ and β .
- Unlike prior conditional normalisation methods, γ and β are not vectors, but tensors with spatial dimensions.
- The produced γ and β are multiplied and added to the normalised activation element-wise.

The generator contains a series of the SPADE residual blocks with upsampling layers



Source: Semantic Image Synthesis With Spatially-Adaptive Normalization

Slide source: [Park et al., 2019]

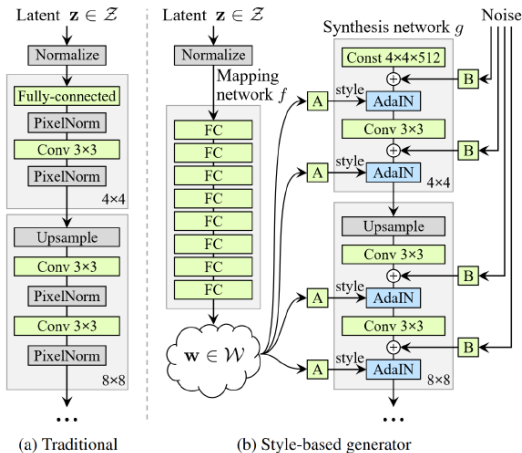


Source: Semantic Image Synthesis With Spatially-Adaptive Normalization

Try yourself: <http://nvidia-research-mingyuliu.com/gaugan>

slide source: [Park et al., 2019]

Applications – A Style-Based Generator Architecture for Generative Adversarial Networks



Source: A Style-Based Generator Architecture for Generative Adversarial Networks [Karras et al., 2019]



Source: A Style-Based Generator Architecture for Generative Adversarial Networks [Karras et al., 2019]

Generative Models II

Case Study

Estimating human gaze:

- Appearance-based gaze estimation
- Eye Contact Detection



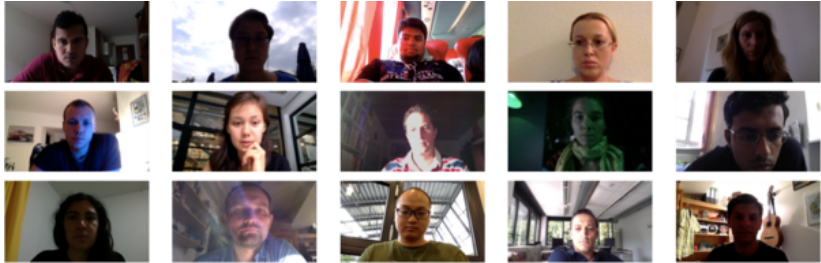
Source: Appearance-based gaze estimation (left), eye contact detection (right) [Băce et al., 2019]

Slide source: [Băce et al., 2019]

(a few) limitations:

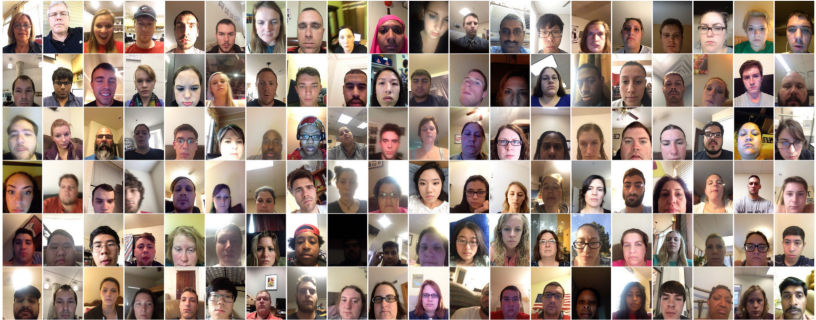
- Difficult to collect ground truth data - there is a need for special-purpose eye tracking equipment
- Need to cover different conditions: illumination, head pose angles, or gaze angles

Slide source: [Băce et al., 2019]



Source: Appearance-based gaze estimation in the wild

Slide source: [Zhang et al., 2015]



Source: Eye Tracking for Everyone

Slide source: [Krafka et al., 2016]



Source: A Large Scale Dataset for Gaze Estimation under Extreme Head Pose and Gaze Variation

Slide source: [Zhang et al., 2020]

One possible solution: Gaze redirection as a data augmentation technique



Source: Photo-Realistic Monocular Gaze Redirection Using Generative Adversarial Networks

Slide source: [He et al., 2019]

- Given an input image x_r , learn G that can redirect the gaze direction in an image into any direction
- $x_r \in \mathcal{R}^{H \times W \times 3}$ and $d_g = [\phi_g, \theta_g]$
- ϕ_g, θ_g are the yaw and pitch angles of the target gaze direction
- Two requirements:
 - Images need to look real and consistent
 - The (new) gaze direction should indicate in the direction specified by the target gaze



Machine Perception and Learning – WS 2025/2026

- The first GAN-based method for gaze redirection using monocular images
- Novel discriminator with dual purpose:
 - Realistic images, as common in many GANs
 - Gaze regularization
- One of the first works to show the potential for augmenting real training datasets with generated images

Adversarial Loss:

$$\mathcal{L}_{adv} = E_{x_r \sim p_{x_r}(x)} [D_{adv}(x_r)] - D_{adv}(G(x_r, d_g)) + \\ \lambda_{gp} E_{\hat{x} \sim p_{\hat{x}}(\hat{x})} [(||\nabla_{\hat{x}} D_{adv}(\hat{x})||_2 - 1)^2]$$

Gaze Estimation Loss:

$$\mathcal{L}_{gaze}^D = E_{x_r \sim p_{x_r}(x)} ||d_r - D_{gaze}(x_r)||_2^2$$

$$\mathcal{L}_{gaze}^G = E_{x_r \sim p_{x_r}(x)} ||d_g - D_{gaze}(G(x_r, d_g))||_2^2$$

Reconstruction Loss:

$$x_{rec} = G(G(x_r, d_g), d_r)$$

$$\mathcal{L}_{rec} = E_{x_r \sim p_{x_r}(x)} ||x_r - x_{rec}||_1$$

Perceptual Loss:

$$\mathcal{L}_C = E_{x_r \sim p_{x_r}(x)} \left[\frac{1}{H_j W_j C_j} \|\psi_j(G(x_r, d_g)) - \psi_j(x_t)\|^2 \right]$$

$$\mathcal{L}_S = E_{x_r \sim p_{x_r}(x)} \left[\sum_{j=1}^J \|f_j(G(x_r, d_g)) - f_j(x_t)\|^2 \right]$$

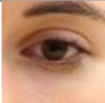
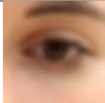
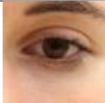
where ψ_j is the j -th activation layer from the pre-trained VGG-16 network and $f_i(x_t)$ is the Gram matrix

Overall Loss:

$$\mathcal{L}_G = -\mathcal{L}_{adv} + \lambda_p \mathcal{L}_p + \lambda_{gaze} \mathcal{L}_{gaze}^G + \lambda_{rec} \mathcal{L}_{rec}$$

$$\mathcal{L}_D = \mathcal{L}_{adv} + \lambda_{gaze} \mathcal{L}_{gaze}^D$$

The perceptual loss was inspired by prior work [Johnson et al., 2016]

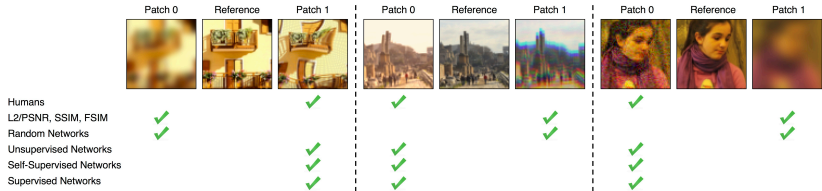
(a) Eye patch	(b) Blurred	(c) Noisy	(d) Shifted
			
MSE	69.57	155.36	176.06
LPIPS	0.122	0.106	0.016

Source: Photo-Realistic Monocular Gaze Redirection Using Generative Adversarial Networks [He et al., 2019]

E.g. image degradations: (a) Eye patch from training set, (b) Blurred with Gaussian filter, (c) With random Gaussian noise, (d) Shifted up by one pixel.

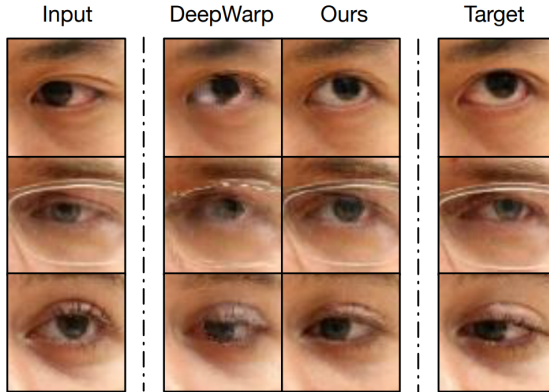
slide source: [He et al., 2019]

- Learned Perceptual Image Patch Similarity (LPIPS)
- Resemble human perception in image evaluation tasks
- Implemented through deep neural nets



Source: The Unreasonable Effectiveness of Deep Features as a Perceptual Metric [Zhang et al., 2018]

Slide source: [Zhang et al., 2018]



Source: Photo-Realistic Monocular Gaze Redirection Using Generative Adversarial Networks [He et al., 2019]

Slide source: [He et al., 2019]

VAEs

- Relatively easier to train
- Explicit inference network $q(z|x)$
- More blurry images (due to reconstruction)

GANs

- Requires many optimization tricks (prone to mode collapse, adversarial objective)
- Implicit generative model (unless using bidirectional GAN)
- Sharper images (due to discriminator)

- If you want to read more about many, many type of GANs:
<https://github.com/hindupuravinash/the-gan-zoo>
- For tips and tricks:
<https://github.com/soumith/ganhacks>

Generative Models III

- Diffusion Models: Discrete Time
- Diffusion Models: Continuous Time
- Diffusion Model Applications

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- J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2223–2232, 2017.